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Original Research

Socio-economic inequality and healthcare costs over the life course – A dynamic microsimulation approach

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ABSTRACT

Objectives: Although there is well-established evidence for the existence of socio-economic inequalities in virtually all dimensions of health, little is known about the implications of these socio-economic disparities for healthcare costs from a cumulative life course perspective. Accounting for differentials in healthcare use patterns and mortality, we assess how lifetime costs differ between socio-economic groups.
Study design: This study used dynamic microsimulation modelling.
Methods: Combining price weights for healthcare services with information on healthcare consumption from the 2014 Austrian Health Interview Survey (n = 15,771), we calculated average cost profiles by gender, age and education consistent with aggregate System of Health Accounts. A dynamic microsimulation model was used to project cumulative healthcare costs over the entire lifecycle for the 2019 birth cohort in four different scenarios to illustrate the influence of the different cost determinants on lifetime costs.
Results: Before considering social inequalities in mortality, men with compulsory education have close to 66% higher lifetime costs than men with tertiary education; for women, the difference is close to 20%. Accounting for longevity differentials reduces this gap to approximately 40% for men and 10% for women. Closing the gap in healthcare use and in longevity between socio-economic groups would reduce lifetime healthcare expenditure by 4.1% in the 2019 birth cohort and by 19% in the whole population.
Conclusions: Accounting for mortality differentials between socio-economic groups has a large impact on estimated lifetime healthcare costs. Reducing social inequalities in health can contribute to containing healthcare expenditures against the backdrop of rising life expectancy.
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Introduction

There is well-established evidence for the existence of socio-economic inequalities in virtually all dimensions of health.[1–3] Little is known about the implications of these socio-economic disparities for healthcare costs, especially from a cumulative life course perspective. Differences in health status by socio-economic group do not necessarily translate into corresponding differences in lifetime healthcare costs for two reasons. First, socio-economic status impacts access to and use of healthcare services, with substantial variation by country and type of service.[4–7] Second, pronounced socio-economic inequalities in life expectancy affect the lifetime use of the healthcare system and the resulting cumulative costs.

A marked social gradient has been observed concerning specialist doctor visits (particularly dentists) and the utilisation of preventive services.[1,4,8,9] In contrast, in most industrialised countries, needs-adjusted social inequalities are much lower for general practitioner (GP) visits and not significant for hospitalisations. Available studies indicate that socio-economic inequalities in health status translate into higher healthcare costs for more vulnerable social groups, at least within health systems that provide universal or nearly universal publicly financed coverage for health services. A recent analysis for the Netherlands shows that total healthcare expenditure for people with lower socio-economic status is between 50% and 150% higher than for people with high socio-economic status, depending on the age group.[10] These results

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are consistent with results from other studies for England, Sweden and Canada.[11–14]

These studies, however, analyse healthcare costs at specific points in time and may thus overstate the extent to which more vulnerable socio-economic groups use the health system over their lifetime. From a life course perspective, the lower life expectancy of these groups tends to offset the higher costs associated with their lower health status. In contrast, earlier onset of diseases and chronic conditions may shift the healthcare costs to younger ages. The extent to which these differential effects impact the social gradient in lifetime health costs must ultimately be determined empirically. This question is of interest to improve our understanding of the distributional effects associated with the healthcare system and gain a fuller picture of economic costs resulting from social inequalities in health.

There is very limited evidence on the link between socio-economic inequality and healthcare costs over the life course. Asaria et al.[15] investigate lifetime hospital costs by level of neighbourhood deprivation in England. Combining data on hospital admissions by age, sex and place of residence with average costs and data on survival, they find that people living in the most affluent areas had lower lifetime hospital costs than those living in more deprived areas. This pattern was more pronounced for women than men and emergency admissions than elective admissions.

We examine lifetime healthcare costs by socio-economic status in Austria using a novel approach that combines survey data and aggregate information from register data to model the healthcare costs with a dynamic microsimulation model. This approach allows us to provide estimates that are consistent with the System of Health Accounts (SHA)[16] for aggregate health costs and with official demographic projections while isolating the effects resulting from different cost determinants, most notably age-related healthcare use patterns, socio-economic differences in mortality and increases in life expectancy. In Austria, health insurance coverage is nearly universal (99.9% in 2018), granting access to a wide range of services and ensuring a comparatively low level of unmet medical needs.[17,18]

We approximate socio-economic status with the educational level, which has a particularly strong correlation with health and is also associated with large differences in mortality.[19,20] On average, across Organisation for Economic Co-operation and Development (OECD) countries, people without a high-school diploma have a life expectancy 6 years shorter than those with tertiary education; the difference is more pronounced for men than for women.[7,18]

Methods

Data on healthcare consumption and healthcare costs

Information on the use of healthcare services was gathered using microdata from the Austrian Health Survey (ATHIS) for 2014. This representative population survey, covering all age groups (N = 15,771), documents the number of inpatient hospital stays (except for stays related to childbirth), daycare stays, as well as the number of visits to GPs and specialist doctors (including hospital outpatient visits). Respondents' educational attainment is coded according to the International Standard Classification of Education (ISCED).

Statistics Austria provides official statistics for aggregate healthcare spending in 2014 by gender and 5-year age groups. These data follow the SHA classification covering the healthcare functions HC.1 to HC.5, representing total healthcare costs for services and goods excluding investments. Our analysis focused on inpatient, outpatient and daycare services. Among these categories, inpatient services are the largest cost item (61%), followed by outpatient services (37%) and daycare services (<2%). Together these services account for 90% of personal healthcare service costs and 71% of total expenditure according to SHA healthcare functions HC.1 to HC.5. In Austria, close to 90% of costs for hospital care and almost 80% of outpatient medical care are covered by government and compulsory insurance schemes.[7] Because hospital stays related to childbirth are not covered by the ATHIS survey, we detract the aggregate childbirth inpatient costs from the healthcare spending data of women in the corresponding age groups. We calculated average healthcare costs by gender and age group using population data.

Based on the information provided by the Austrian Ministry for Work, Social Affairs, Health and Consumer Protection and by the Austrian National Public Health Institute, we determined price weights for the different healthcare service categories under scrutiny. Inpatient hospitalisations have by far the highest average unit cost (856 Euro per day) and are the most important determining factor in cost estimation. GP and specialist doctor visits were assigned a price weight of 57 Euro and 76 Euro, respectively. For daycare, which is not used very often and for which no price reference was available, we assumed a unit cost of 600 Euro.

Estimation of cost profiles

In the first step, we analysed the use of healthcare services with the ATHIS data, calculating how the consumption of doctor and hospital visits is distributed by gender, age and education. We distinguished three levels of education: low (at most compulsory schooling, ISCED 0–2), medium (lower and upper secondary education and apprenticeship, ISCED 3–4) and high (tertiary education, ISCED 5+). The disaggregation by education level was limited to ages 20–85 years: For the under-20-year-olds, a differentiation according to education categories is not meaningful. For ages >85 years, differentiation by education level would be problematic because of the low number of observations in some groups, particularly women with higher education. Although this is a limitation, previous findings indicate that socio-economic differences in health tend to decrease at higher ages.[21–23] Thus, for individuals aged < 20 years and those aged >85 years, healthcare utilisation was assumed to be independent of educational attainment.

In the second step, we used the healthcare price weights to determine individual healthcare costs. To ensure correspondence to the average SHA expenditure profiles by gender and age group, we calibrated the average healthcare costs by gender, age group and education.

Microsimulation of costs over the life course

To calculate healthcare expenditure over the entire lifecycle, we used the dynamic microsimulation model microWELT.[24] Based on European Union Statistics on Income and Living Conditions data, microWELT provides a representative cross-section of the Austrian population regarding gender, age and education structure in the baseline year. The model consists of several interconnected family demographic, health and socio-economic modules and models the further development of the Austrian population.

Following Asaria et al.,[15] we assumed that the healthcare cost profiles by gender, age and education remain constant. We estimated cumulative lifetime costs by combining these profiles with mortality information for different socio-economic groups. Each individual in the microWELT population was attributed the average healthcare costs according to gender, age and education. To apply gender- and education-specific survival probabilities for each age, we combined data on remaining life expectancy by education for 25- and 65-year-olds[25] with the actuarial life tables.[26] The model

reflects population projections by Statistics Austria by reproducing changing age-specific mortality rates and accounts for the overall increase in life expectancy. Relative differences in mortality between education groups were kept constant throughout the simulation.

Our approach allowed for calculating yearly healthcare costs for the total population consistent with SHA aggregates and determining lifetime costs for a specific population cohort. We focused on the cohort of newborns in 2019, summing the costs over ages to obtain the cumulative lifetime costs by gender and education group. Exploiting the flexibility of our model, we separated and quantified the effects that different assumptions, such as changes in life expectancy, have on cumulative lifetime costs. The analyses were performed using the open-source software R (version 4.2.2) and the open-source dynamic microsimulation programming technology Modgen.[27] All codes and the model can be downloaded from the project website www.microWELT.eu.

Results

Healthcare cost profiles by gender, age and education

Fig. 1 shows the cost profiles for men and women by age and education. While the cost profiles increase significantly with age, the costs of persons with higher education tend to be lower than those of the other education groups at all ages. While women show larger differences by education above 40, men display larger differences in younger groups. According to the calculations provided by Statistics Austria, there is a large gap in absolute expenditure levels by gender in the highest age group, with women aged ≥90 years reporting, on average, 57% higher costs than men. This large gap can be traced back primarily to inpatient services, where average expenditures are approximately 17,000 Euro for women but only close to 11,000 Euro for men. This sizeable difference can probably be explained by hospitalisations connected to long-term care needs, which affect women more than men.

Assuming constant age-specific mortality and unchanged relative differences in mortality between education groups, the life expectancy of men born in 2019 ranges from 76 years (low education) to 83 years (higher education). In contrast, women show smaller education-specific differences, with life expectancy varying between 83 years (low education) and 86 years (higher education). Fig. 2 shows the corresponding survival curves. A steeper educational gradient in mortality for men than for women has been documented in previous studies for European countries as well as for the United States.[28,29] The causes for this gender difference, however, are not yet fully understood and might include, *inter alia*, behavioural factors and differences in the distribution across hazardous workplaces by gender and education.

When considering forecasts of increasing life expectancy, the average life expectancy of those born in 2019 ranges between 85 and 93 years, depending on education and gender (see Figure A1 in Appendix 1).

Healthcare cohort cost profiles

The microsimulation projection combines data on survival and healthcare costs by gender, age and education. Fig. 3 displays the age-specific costs for the 2019 birth cohort with constant age-specific mortality. The data reflect average expenditure levels per cohort member, that is, costs at each age applied to the share of individuals of the birth cohort still alive. For both genders, at older ages, the combination of different survival probabilities and age-specific costs leads to decreasing average costs in all three education groups.

Socio-economic differences in mortality lead to a reversal of the cost ratios between education groups along the age dimension. Those with lower education have higher average health costs at younger ages but lower costs at older ages. The pattern is less pronounced for women, where expected healthcare expenditures for those with lower education are slightly higher than for the other

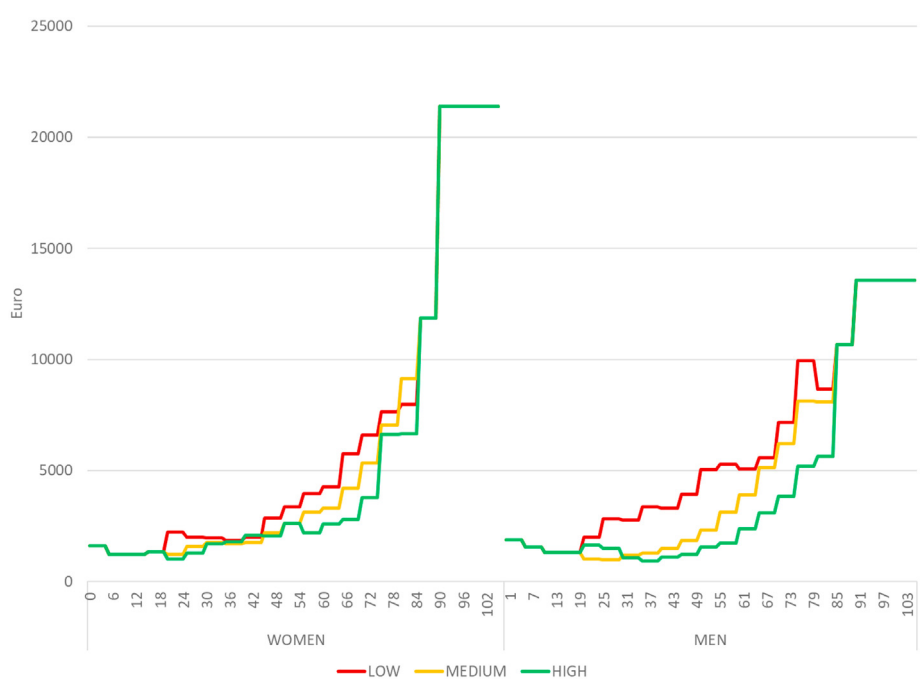

Fig. 1. Age profile of public health expenditures by gender and education. On average, the expenditure levels correspond to official statistics for public healthcare spending in 2014 covering inpatient, outpatient and daycare services, provided by Statistics Austria by gender and age (in 5-year groups) following the System of Health Accounts (SHA) classification. 2019 prices.

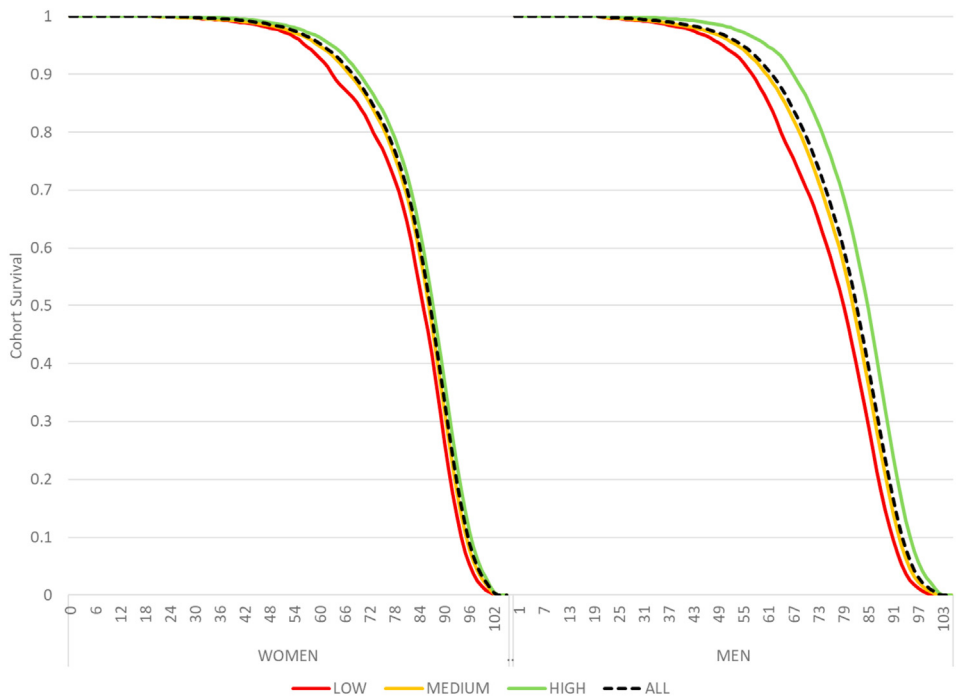

Fig. 2. Survival curves without mortality improvements. Survival curves for the 2019 birth cohort based on remaining life expectancy by education for 25- and 65-year-olds by Murtin et al. and actuarial life tables provided by Statistics Austria (2019), assuming constant age-specific mortality.

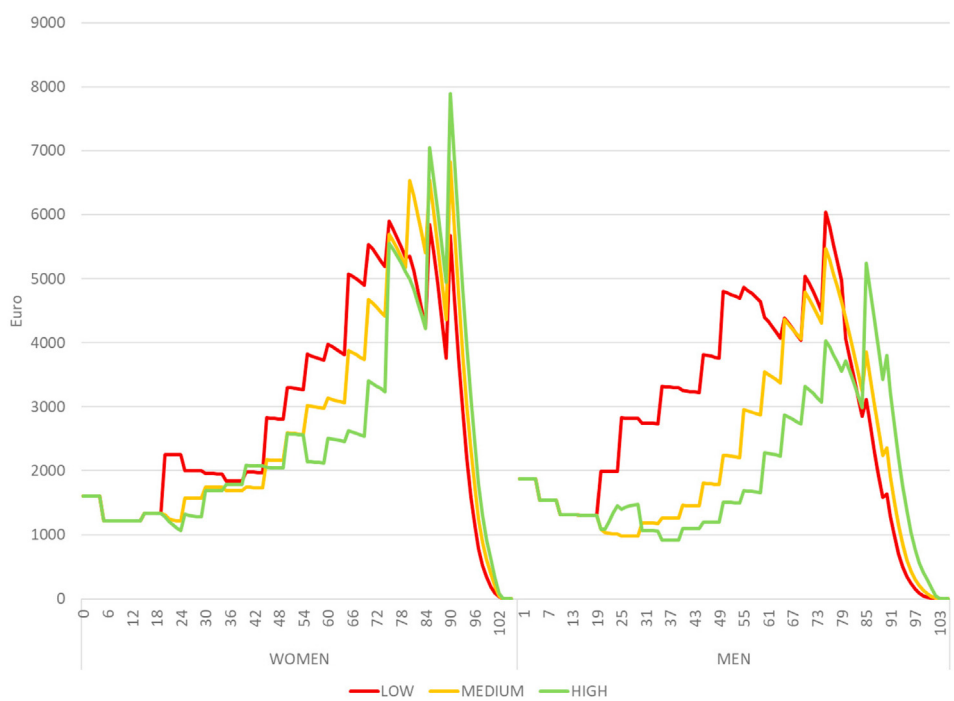

Fig. 3. Cohort profile of healthcare expenditures. Microsimulation results for the 2019 birth cohort combining healthcare costs by gender, age and education (Fig. 1) and survival curves by gender and education (Fig. 2). The data reflect average expenditure levels per member of the birth cohort, assuming constant age-specific mortality. 2019 prices.

education groups up to age 75 years. For men, those with low education have significantly higher expenditure levels between the ages of 20 and 65 years. After that point, higher mortality in the lower education group leads costs first to converge to the level of those with medium education and then to fall significantly below the level for men with high education.

Figure A2 in Appendix 1 shows the results for the cohort cost profiles taking into account increasing life expectancy to reflect population projections. The cost curves shift to the right, with higher expenditures at older ages. The increase in healthcare costs is particularly strong for men aged >70 years and women aged >80 years.

Life course healthcare costs by gender and education

Total healthcare costs over the life course by gender and education result from the sum of the age-specific per-capita values in Fig. 3. To illustrate the influence of the different cost determinants, Table 1 presents the results in four steps. The first column shows what the healthcare costs of the cohort born in 2019 would be if life expectancy remained unchanged and if there were no differences in life expectancy by education group. In this scenario, the lifetime costs of men with lower education are >80% higher than those with high education, and for women, the difference amounts to 25%. The second column displays the results for constant life expectancy but considers education-specific differences in mortality. Here, the differences in costs across education groups are reduced to approximately 50% for men and 13% for women.

The third and fourth columns present the results of scenarios where increasing life expectancy, consistent with the official population projections, is considered. The third column displays calculations neglecting differences in education-specific mortality, whereas the fourth column incorporates the social gradient in life expectancy. In both cases, lifetime healthcare costs rise by nearly 40% for women and approximately 35% for men. Differences between education groups are smaller than in the first two scenarios, where life expectancy was kept constant. Higher life expectancy increases lifetime healthcare costs for all groups, thus reducing the relative impact of socio-economic differences. Without education-specific differences in mortality, the healthcare costs of people with low educational attainment are close to 66% higher for men and 20% higher for women with high education. Including education-specific life expectancy in the model narrows the differences to 40% and 10%, respectively. Shorter life expectancy accounts for a reduction in lifetime costs of the lower education group by approximately 8% for men and 6% for women. On the other hand, lower mortality leads to an increase in the estimated lifetime healthcare costs of the higher education group by about 8% for men and 3% for women.

Whether we consider increasing life expectancy or not, considering the social gradient in mortality reduces the expenditure gap across education groups by approximately 40% for men and almost 50% for women.

Discussion

Main findings

In line with previous research, our estimates indicate that in a longitudinal perspective, lower life expectancy reduces but does not outweigh the higher average healthcare costs that we observe for vulnerable socio-economic groups from a cross-sectional perspective.[15] Using education as an indicator of socio-economic status, we find that inequalities in healthcare expenditure are larger for men than for women. Before accounting for the social gradient in mortality, the lifetime costs of men and women with compulsory education are about two-thirds and one-fifth higher, respectively, than those of men and women with tertiary education. Taking education-specific differences in mortality into account, which are more pronounced for men than for women, this difference is reduced to approximately 40% for men and 10% for women. These values refer to projections for the cohort born in 2019 and include an increase in life expectancy consistent with population projections.

If the population groups with lower and medium education had the sex- and age-specific healthcare costs associated with those with higher education, the total healthcare expenditure for the 2019 birth cohort would be approximately 4.1% lower. This difference can be interpreted as the low long-term benchmark for the costs associated with socio-economic inequalities in health, as the 2019 cohort – with a projected share of 38% with high and below 10% with low education – has a far more favourable educational composition than the current population. In contrast, current health expenditures would be 19% lower if the average age-specific costs of the high education group would apply to the whole population. While out of the scope of this article, our microsimulation approach is specifically powerful in projecting the evolution of period costs under various scenarios.

Strengths and limitations

Our study contributes to filling the research gap concerning the link between socio-economic inequality and healthcare system use from a life course perspective. The results provide an interesting benchmark for other countries with a comprehensive public healthcare system. Using a dynamic microsimulation model in this context is novel and has several advantages. It produces results that are consistent with aggregate SHA statistics as well as with the official population projections. Covering different healthcare services, we can account for a large share of expenditures in the healthcare system. The method allows isolating effects resulting from different cost determinants, most notably age-related healthcare use patterns, socio-economic differences in life expectancy and increasing life expectancy.

These strengths notwithstanding our study are subject to several limitations, which will have to be addressed in future research. First, for our cohort analysis and lifetime projections, we

Table 1
Lifetime healthcare costs by gender and education.

Gender and education		Constant age-specific mortality; no differential mortality by education	Constant age-specific mortality; differential mortality by education	Decreasing mortality; no differential mortality by education	Decreasing mortality; differential mortality by education
		In Euro per person (in 2019 prices)			
Female	Low	318,375	298,010	430,535	406,092
	Medium	284,084	279,038	395,095	389,068
	High	253,602	262,478	359,492	369,443
	All	274,462	273,710	383,368	382,532
Male	Low	324,712	295,842	414,295	379,937
	Medium	232,284	224,706	312,595	304,810
	High	176,683	197,951	249,856	270,006
	All	222,703	222,054	301,514	300,261
Total	Low	321,835	296,795	421,493	391,138
	Medium	256,173	249,702	350,629	343,823
	High	220,778	235,095	312,762	327,050
	All	248,455	247,702	342,211	341,152

Total healthcare costs over the life course by gender and education resulting from the sum of the age-specific per-capita values in Fig. 3. Costs based on 2014 SHA figures by Statistics Austria.

had to rely on the simplifying assumption that the age profiles of healthcare costs remain constant and can be used for cohort analysis. Second, we did not control for needs-adjusted inequalities in healthcare use. If disadvantaged social groups face barriers in accessing the healthcare system or are less likely than more advantaged groups to seek healthcare services when they need them, our results (that are based on data for actual use) might underestimate the impact of socio-economic health differences. Available studies indicate that socio-economic inequalities in healthcare use in Austria are pronounced concerning specialist doctor visits, but that hospital stays (which account for the largest share of health costs) and GP visits are equitably distributed.[4,30] Third, we did not account for end-of-life healthcare costs. A study for England shows that end-of-life healthcare costs are higher among patients with lower socio-economic status, even after being controlled for patient-level characteristics.[31] In this respect, too, our projections may underestimate social inequalities in using the health system. Finally, in our study, we use educational attainment to approximate socio-economic status without investigating the mechanisms that determine the relationship between education, health and mortality. As highlighted by the gender difference that we observe in the educational mortality gradient, educational attainment may be operating differently across population subgroups to influence health and mortality.[32] To gain a more comprehensive picture of the impact that socio-economic inequalities have on lifetime healthcare costs, more studies using different indicators for socio-economic status will be needed.

Author statements

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Ethical approval

No ethical approval was applicable. This study did not involve human participants or animal subjects. The manuscript does not contain personal and/or medical information about an identifiable individual and/or a case report/case history.

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Competing interests

None declared.

Data availability

Microdata used for this research cannot be made available because of legal restrictions: Microdata from the Austrian Health Survey (ATHIS) and the European Union Statistics on Income and Living Conditions (EU-SILC) can be requested free of charge from Statistics Austria for scientific analysis by members of a university or a relevant scientific institution (email: forschungundlehre@statistik.gv.at). Users are not allowed to pass on the data to third parties. (http://www.statistik.at/web_de/services/mikrodaten_fuer_forschung_und_lehre/datenzugang/antrag_fuer_sds/index.html). All codes and the model can be downloaded from https://www.microwelt.eu/Model/Model-Index.html.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.puhe.2023.04.001.

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